IT INFRASTRUCTURE AND COMPUTER NETWORKS

Week 7

IP ADDRESSES

172.16.10.9

2.4.3.7

172.169.10.67

10.10.4.2

192.168.14.5

8.8.8.8

203.123.45.2

192.168.1.254

255.255.255.0

Binary to Decimal and vice versa

1	1	0	1	1	0	0	1

$1 \times 2^0 = 1 \times 1 = 1$

$0 \times 2^1 = 0 \times 2 = 0$

$0 \times 2^2 = 0 \times 4 = 0$

$1 \times 2^3 = 1 \times 8 = 8$

$1 \times 2^4 = 1 \times 16 = 16$

$0 \times 2^5 = 0 \times 32 = 0$

$1 \times 2^6 = 1 \times 64 = 64$

$1 \times 2^7 = 1 \times 128 = 128$

$1 + 8 + 16 + 64 + 128 = 217$

CONVERT THE DECIMAL 78 TO BINARY

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$78 = 64 + \underbrace{14}$

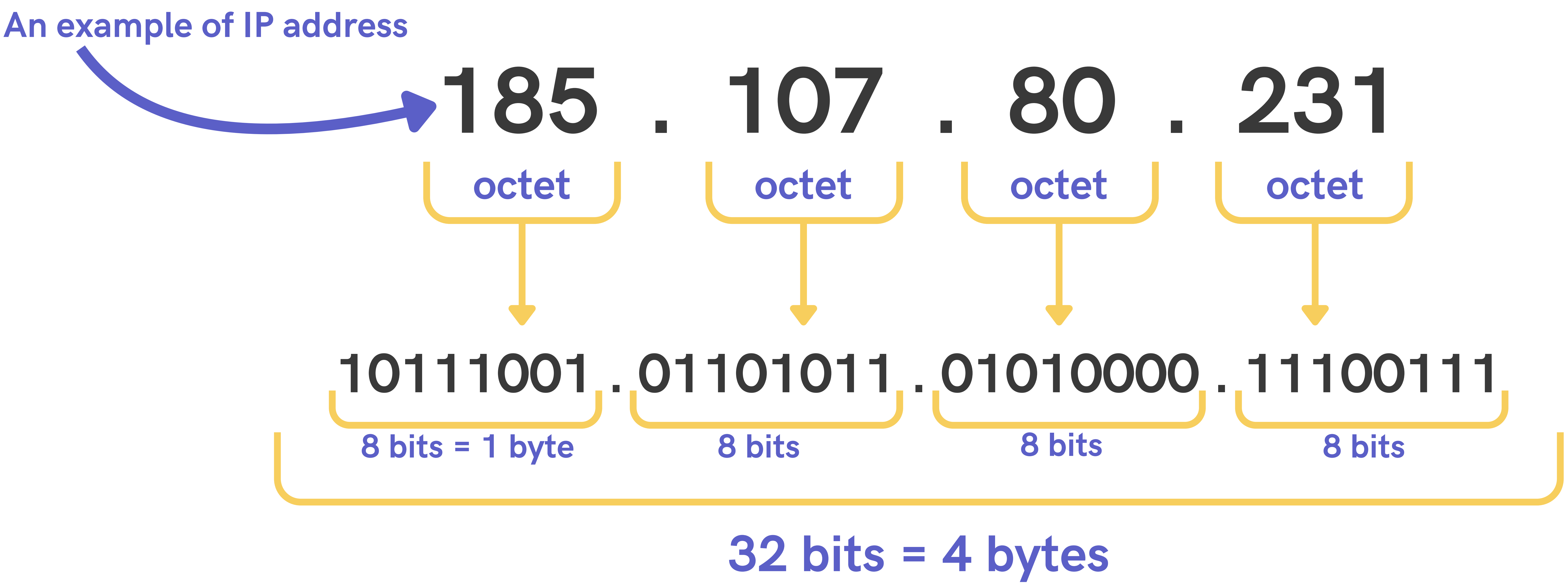
$14 = 8 + \underbrace{6}$

$6 = 4 + 2$

1 0 0 1 1 1 0

What is IP address?

4 dot-separated numbers in range from 0 to 255



What is subnet mask?

32-bit number that accompanies an IP address to divide it into network and host components

An example of subnet mask

there are 24 “1”s in a row

255.255.255.0 or just /24

11111111.11111111.11111111.00000000

Subnet Mask	CIDR	Subnet Mask	CIDR
255.128.0.0	/9	255.255.240.0	/20
255.192.0.0	/10	255.255.248.0	/21
255.224.0.0	/11	255.255.252.0	/22
255.240.0.0	/12	255.255.254.0	/23
255.248.0.0	/13	255.255.255.0	/24
255.252.0.0	/14	255.255.255.128	/25
255.254.0.0	/15	255.255.255.192	/26
255.255.0.0	/16	255.255.255.224	/27
255.255.128.0	/17	255.255.255.240	/28
255.255.192.0	/18	255.255.255.248	/29
255.255.224.0	/19	255.255.255.252	/30

For example:

192.168.10.2/24

IP address: 11000000.10101000.00001010 00000010

Subnet mask: 11111111.11111111.11111111 00000000

Network portion

Host portion

3 important addresses:

- Network Address - 1st address - only 0s in Host portion
- Broadcast Address - last address - only 1s in Host Portion
- Host Address - everything between NA and BA
    - First Host Address - after NA - all 0s and last 1 in Host portion
    - Last Host Address - before BA - all 1s and last 0 in Host portion

So, what are the NA, FHA, LHA and BA for 192.168.10.2/24?

- NA: 192.168.10.0/24
- FHA: 192.168.10.1/24
- LHA: 192.168.10.254/24
- BA: 192.168.10.255/24

Task:

172.16.144.85/24

Identify NA, FHA, LHA and BA.

What if we change subnet mask to /16?

Identify NA, FHA, LHA and BA.

What if we change subnet mask to /27?

Identify NA, FHA, LHA and BA.

Calculating hosts in a network

$$\text{Hosts} = 2^n - 2$$

where

n = number of host bits

For example:

subnet mask = /24 ----> host bits = 32 - 24 = 8 ----> $2^8 - 2 = 254$ hosts

Subnetting

The process of logically dividing a single large IP network into smaller sub-networks to improve performance, security, and address management.

FLSM

Fixed Length Subnet Mask

- Equal subnets size
- The same subnet mask
- Often wastes IP addresses
- Outdated

VLSM

Variable Length Subnet Mask

- Different subnets size
- Different subnet masks
- Minimizes IP addresses wasting

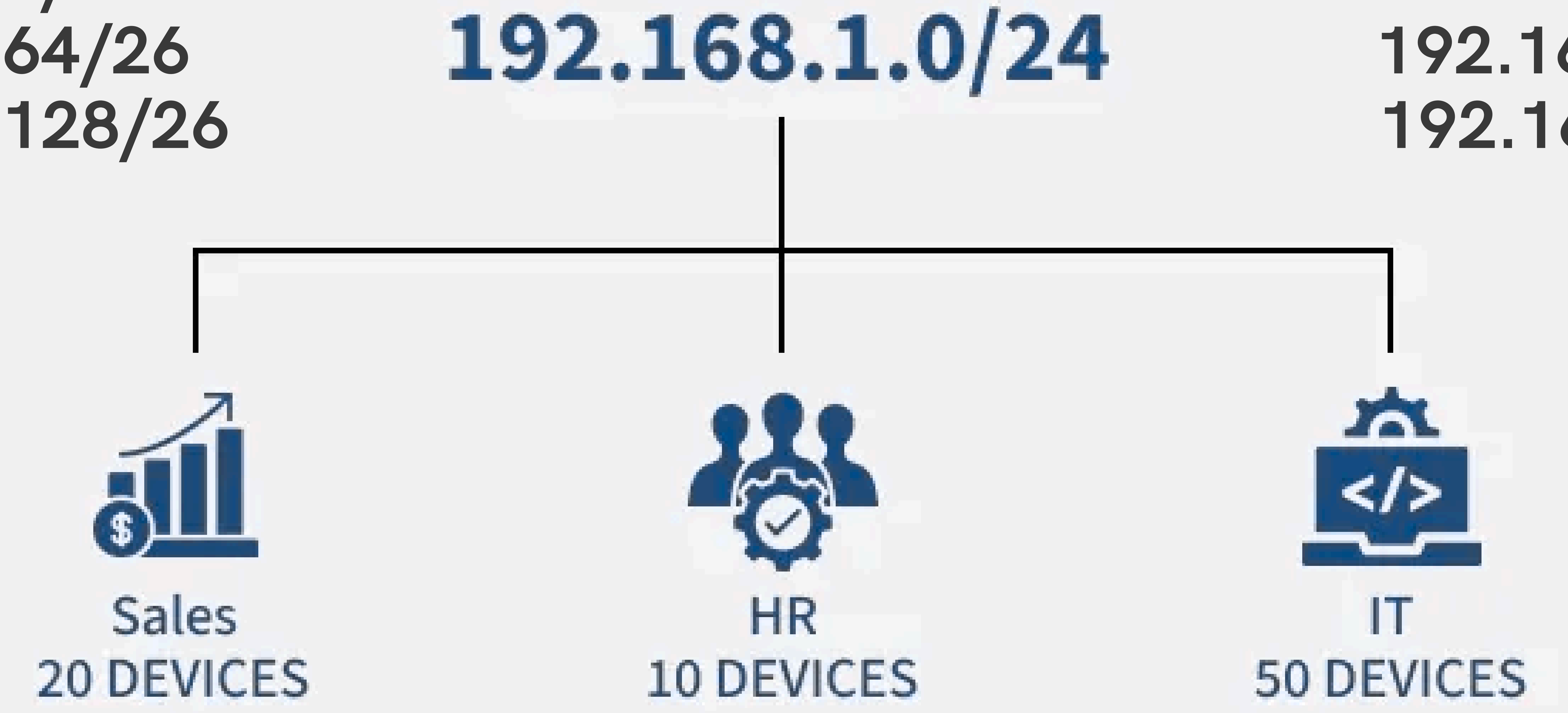
For example:

FLSM

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26

VLSM

192.168.1.0/26
192.168.1.64/27
192.168.1.96/28



How to VLSM?

- Sort future subnets by host amount, from biggest to smallest. Start with the biggest.

IT (50) → Sales (20) → HR(10)

- Find the closest degree of 2 that will fit this amount of hosts. This will define the subnet mask.

2^5? (32 - 2) < 50
2^6? (64 - 2) > 50 → 32 - 6 = 26 → /26

- Assign the first subnet from the beginning of the network.

192.168.1.0/26

- Write the network address, first host, last host, and broadcast address.

NA: 192.168.1.0/26, FHA: 192.168.1.1/26, LHA: 192.168.1.62/26, BA: 192.168.1.63/26

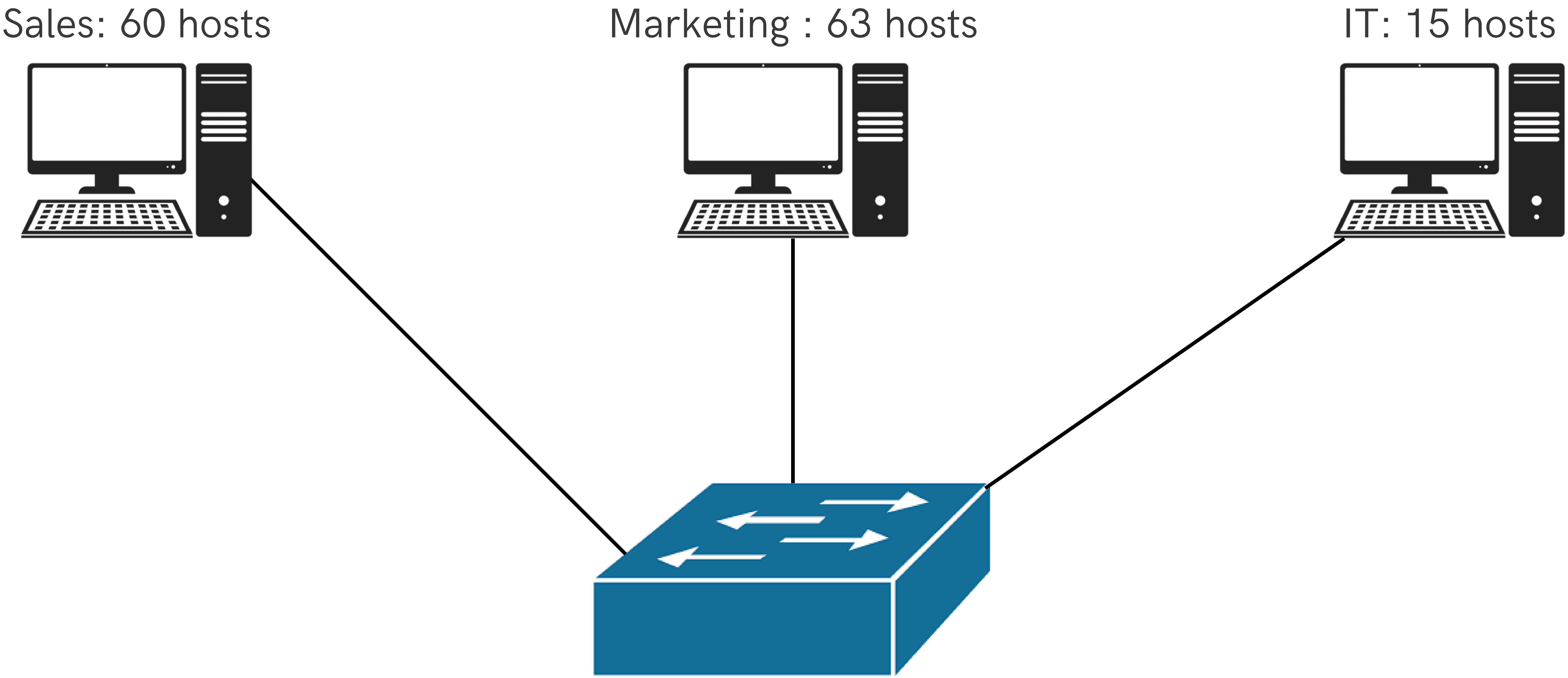
- Move to the next available IP address.
- Repeat the process for all networks.

Magic Number

192.168.1.0/26
=> 32-26=6
=> $2^6 = 64$

64 is a Magic Number

192.168.1.0/26 + 64 = 192.168.1.64
- start of the next subnet



NETWORK: 172.16.30.0/24

TRY YOURSELF

Given Network: 10.10.16.0/20
6 Subnets:
├── Main Office LAN: 500 hosts
├── Branch Office 1: 100 hosts
├── Branch Office 2: 50 hosts
├── Server Farm: 30 hosts
├── DMZ: 10 hosts
└── Management VLAN: 10 hosts

Design VLSM, find network address, host range, broadcast address and default gateway for each subnet

How many IP addresses we have in total?

$$2^{32} = 4\ 294\ 967\ 296$$

Is it enough nowadays?

How to solve this problem?

IPv6

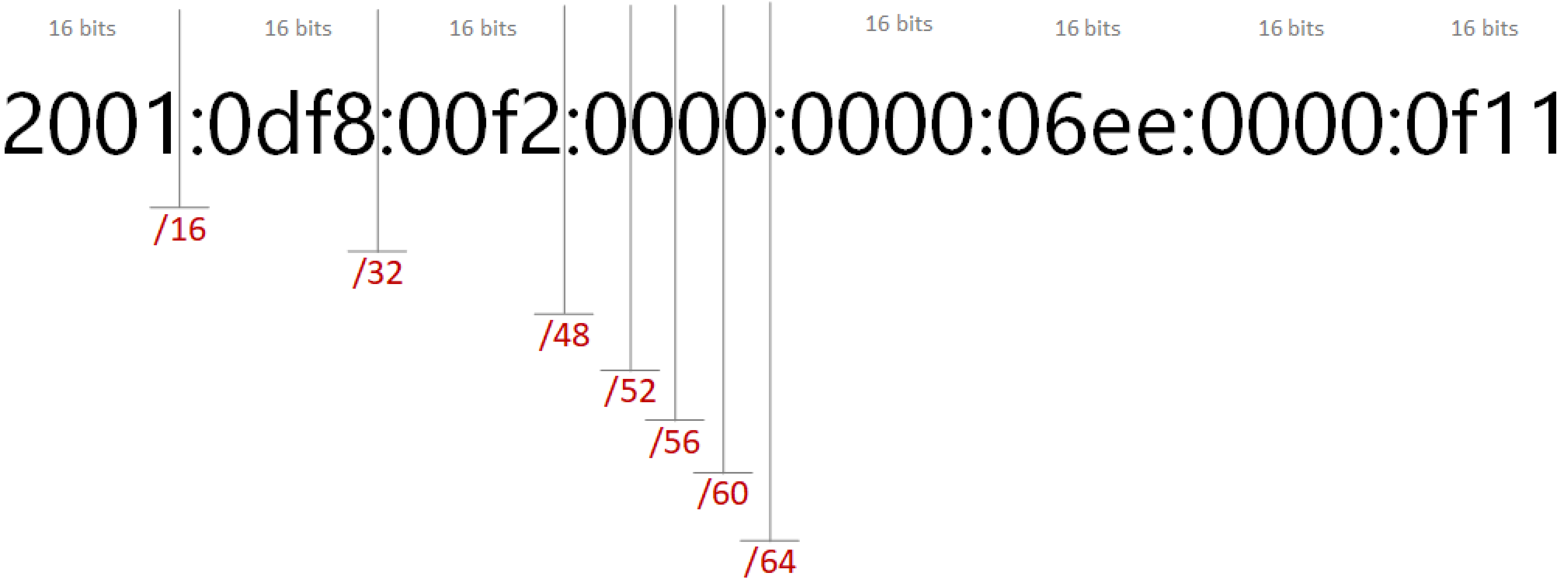
IPv6 address

2001 : 0DC8 : E004 : 0001 : 0000 : 0000 : 0000 : F00A

16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits : 16 bits

128 Bits

IPv6 prefix


16 bits
16 bits
16 bits
16 bits
16 bits
16 bits
16 bits
2001:0df8:00f2:0000:0000:06ee:0000:0f11
/16
/32
/48
/52
/56
/60
/64

How many IPv6 addresses we have in total?

$$2^{128}$$

or

340 undecillion (340 trillion trillion trillion)

or

340 282 366 920 938 463 463 374 607 431 768 211 456

DECIMAL	HEX	BINARY
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

IPv6 shortening

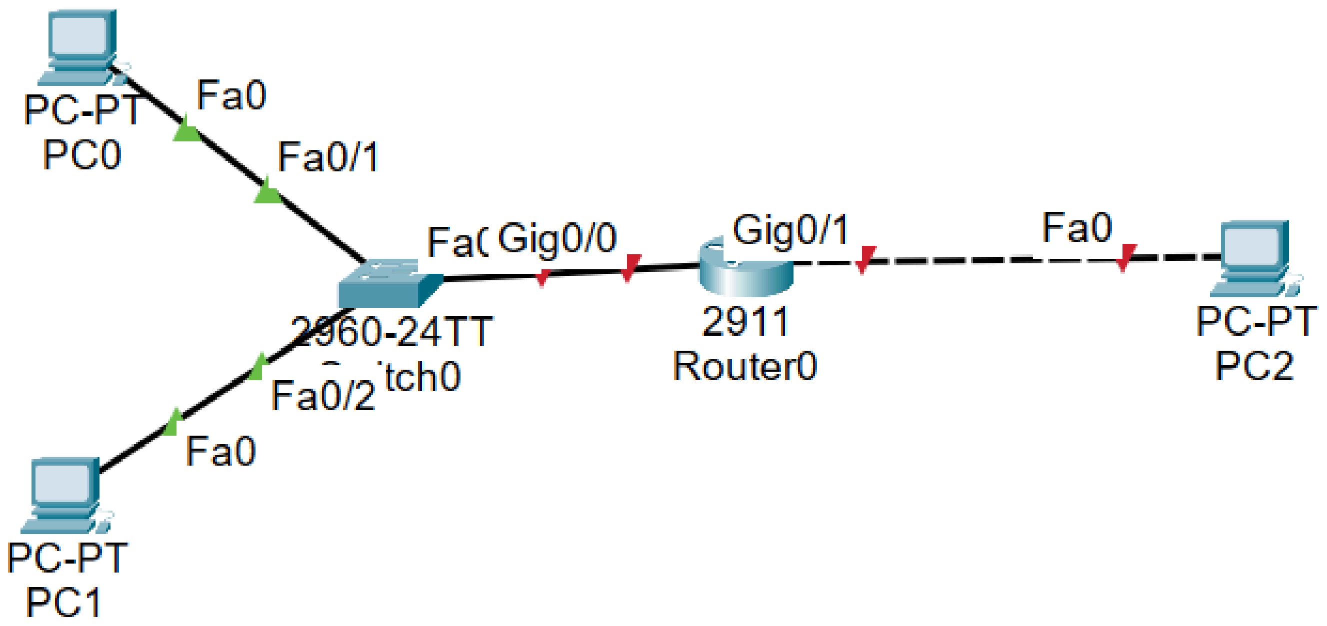
Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8
2001	0df8	00f2	0000	0000	06ee	0000	0f11

2001:0df8:00f2:0000:0000:06ee:0000:0f11

Group 1	Group 2	Group 3	Group 6	Group 7	Group 8
2001	0df8	00f2	06ee	0000	0f11

2001:0df8:00f2::06ee:0000:0f11

IPv6 configuration



IPv6 Configuration

○ Automatic	● Static
IPv6 Address	2001:db8:1::2 / 64
Link Local Address	FE80::201:C9FF:FE03:DACA
Default Gateway	fe80::1

```
Router(config)#int g0/0
Router(config-if)#ipv6 address 2001:db8:1::1/64
Router(config-if)#ipv6 address fe80::1 link-local
Router(config-if)#no shutdown
```

```
C:\>ping 2001:DB8:2::2

Pinging 2001:DB8:2::2 with 32 bytes of data:

Reply from 2001:DB8:2::2: bytes=32 time=11ms TTL=127
Reply from 2001:DB8:2::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:2::2: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:2::2: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:2::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

```
Router(config)#ipv6 unicast-routing
```

Any questions?